

Xiaoxia (Shirley) Wu

Senior Staff Scientist

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in [xiaoxiawu1](#)

Google Scholar

Summary

Senior Staff Scientist with 10+ years of experience in machine learning research and systems, specializing in LLM inference optimization, speculative decoding, and model compression. Led production-scale projects at Together AI and Microsoft (DeepSpeed), with 20+ publications at top venues including 4 oral presentations. 6,100+ citations (h-index 24). Ph.D. from UT Austin with the Frank Gerth III Dissertation Award.

Experience

- Jul. 2024 – **Senior Staff Scientist**, *Together AI*, Remote
Present
 - Led **Aurora** and **ATLAS** projects, a unified training-serving system for speculative decoding that reduces training-serving mismatch through continuous online adaptation from live inference traffic.
 - Built and scaled speculator training and distillation pipelines for production LLM serving, spanning supervised fine-tuning, distillation, and RL-driven post-training.
 - Drove full-stack efficiency across quantization (FP8/FP4/INT4/NVFP4), inference optimization, and deployment in vLLM, SGLang, and TensorRT-LLM.
 - Built and led a specialized team spanning model optimization, evaluation, and production inference.
 - Published blogs: [Aurora](#), [ATLAS](#) and [Fastest Inference for DeepSeek-R1 on NVIDIA HGX B200](#).
- Nov. 2021 – **Senior Researcher**, *DeepSpeed Team*, *Microsoft*, Redmond, WA
Jun. 2024
 - Led quantization projects: [DeepSpeed-FP6](#), [ZeroQuant-FP](#), [ZeroQuant-HERO](#), and [ZeroQuant](#), achieving significant inference speedups and memory reductions for large language models.
 - Co-led development of *DeepSpeed-Chat* and *DeepSpeed-VisualChat*, enabling affordable RLHF training of ChatGPT-like models.
 - Published high-impact papers including *Extreme Compression* in [NeurIPS 2022 oral](#)
- Jan. 2021 – **Postdoctoral Researcher**, *University of Chicago*, *Host*: Rebecca Willett
Oct. 2021
 - Developed algorithms for differentially private empirical risk minimization.
 - Conducted theoretical analysis on privacy-utility tradeoffs in machine learning models.
- May 2020 – **Research Intern**, *Google*, Mountain View, CA, *Mentor*: Behnam Neyshabur
Aug. 2020
 - Researched curriculum learning for CIFAR10 and ImageNet.
 - Demonstrated that curriculum learning provides no consistent benefit over standard training; results published as [ICLR 2021 oral](#).
- Aug. 2017 – **Research Intern**, *Meta AI Research*, New York, NY, *Mentor*: Léon Bottou
Oct. 2018
 - Implemented and benchmarked state-of-the-art optimization algorithms in PyTorch.
 - Analyzed optimization of layer and weight normalization; work presented as [ICML 2019 oral](#).

Education

- Aug. 2014 – **Ph.D. in Machine Learning**, *The University of Texas at Austin*
Dec. 2020
 - Advisors: Rachel Ward (supervisor) & Léon Bottou (co-supervisor)
 - Thesis (Frank Gerth III Dissertation Award): *Gradient-based Optimization and implicit regularisation over non-convex landscapes*

Publications and Preprints (selected)

Google Scholar: 6,190 citations, h-index 24, i10-index 30, as of March 17, 2026.

Recent LLM Systems, Reasoning, and Speculative Decoding

- [1] J. Wang, F. Bie, J. Li, Z. Zhou, Z. Shao, Y. Wang, Y. Liu, et al, **X. Wu (project lead)** *When RL Meets Adaptive Speculative Training*. [arXiv:2602.06932](#) (2026).
- [2] Z. Zhou, F. Bie, Z. Chen, Z. Zhang, Y. Yang, J. Wang, B. Athiwaratkun, **X. Wu (project lead)**, S.L. Song. *CARE: Covariance-Aware and Rank-Enhanced Decomposition for Enabling Multi-Head Latent Attention*. [ICLR](#) (2026).

- [3] H. Singh, X. Li, K. Sareen, M. Maheswaran, S. Tan, **X. Wu**, et al. *V1: Unifying Generation and Self-Verification for Parallel Reasoners*. *arXiv:2603.04304* (2026).
- [4] Z. Shao, V. Srivatsa, Q. Wu, A. Ariyak, **X. Wu**, et al. *Beat the Long Tail: Distribution-Aware Speculative Decoding for RL Training*. *arXiv:2511.13841 MLSYS* (2026).
- [5] Z. Zhang, **X. Wu (project lead)**, et al. *Understanding and Steering the Cognitive Behaviors of Reasoning Models at Test-Time*. *arXiv:2512.24574* (2025).

Model Compression

- [1] H. Xia, **X. Wu (project lead)**, et al. *Kitty: Accurate and Efficient 2-bit KV Cache Quantization*. *arXiv:2511.18643 MLSYS* (2026).
- [2] H. Xia, Z. Zheng, **X. Wu**, et al. *FP6-LLM: Efficiently Serving LLMs through FP6-Centric Algorithm–System Co-Design*. *arXiv:2401.14112* (2024).
- [3] **X. Wu**, H. Xia, S. Youn, Z. Zheng, et al. *ZeroQuant(4+2): Redefining LLM Quantization with an FP6 Strategy*. *arXiv:2312.08583* (2023).
- [4] **X. Wu***, Z. Yao*, et al. *ZeroQuant-FP: W4A8 Post-Training Quantization Using Floating-Point Formats*. *NeurIPS ENLSP* (2023).
- [5] G. Wang, H. Qin, S.A. Jacobs, **X. Wu**, et al. *ZeRO++: Extremely Efficient Collective Communication for Large Model Training*. *ICLR* (2024).
- [6] Z. Yao, **X. Wu**, C. Li, S. Youn, Y. He. *ZeroQuant-V2: Exploring Post-Training Quantization in LLMs*. *AAAI* (2024).
- [7] **X. Wu**, C. Li, R.Y. Aminabadi, Z. Yao, Y. He. *Understanding INT4 Quantization for Transformer Models*. *ICML* (2023).
- [8] **X. Wu**, Z. Yao, et al. *XTC: Extreme Compression for Pre-trained Transformers Made Simple and Efficient*. *NeurIPS* (2022), *oral*.
- [9] Z. Yao, R.Y. Aminabadi, M. Zhang, **X. Wu**, et al. *ZeroQuant: Efficient Post-Training Quantization for Transformers*. *NeurIPS* (2022).

Optimization and Theory

- [1] **X. Wu**, E. Dyer, B. Neyshabur. *When Do Curricula Work?* *ICLR* (2021), *oral*.
- [2] **X. Wu**, Y. Xie, et al. *Adaloss: A Computationally-Efficient Adaptive Gradient Method*. *AAAI* (2022).
- [3] **X. Wu***, E. Dobriban*, T. Ren*, et al. *Implicit Regularisation and Convergence for Weight Normalisation*. *NeurIPS* (2020).
- [4] R. Ward*, **X. Wu***, L. Bottou. *Adagrad Stepsizes: Sharp Convergence over Nonconvex Landscapes*. *ICML* (2019), *oral*.

NLP and Multi-modal

- [1] Z. Yao, **X. Wu**, C. Li, M. Zhang, H. Qin, et al. *DeepSpeed-VisualChat: Multi-Round Multi-Image Interleave Chat*. *arXiv:2309.14327* (2023).
- [2] Z. Yao, R.Y. Aminabadi, O. Ruwase, S. Rajbhandari, **X. Wu**, et al. *DeepSpeed-Chat: Affordable RLHF Training of ChatGPT-like Models*. *arXiv:2308.01320* (2023).
- [3] C. Li, Z. Yao, **X. Wu**, M. Zhang, Y. He. *DeepSpeed Data Efficiency: Improving Deep Learning Model Quality*. *AAAI* (2024).

Honours and Awards

- 2020 Frank Gerth III Dissertation Award (top dissertation in Mathematics, UTAustin)
- 2019 ICML & NeurIPS Travel Awards
- 2018, 2019 Professional Development Award, UTAustin
- 2018 Graduate School Fellowship, UTAustin
- 2012 Scotland Saltire Scholarship, University of Edinburgh

Professional Service & Mentorship

- Journal Reviewer: *Journal of Machine Learning Research*
- Conference Reviewer: NeurIPS, ICML, ICLR, AISTATS, MSML, WiML
- Mentor: Directed Reading Program, UTAustin
- Volunteer Teacher: Sanger Center, UTAustin